Chapter C2: Chemical patterns

Overview

This chapter features a central theme of modern chemistry: it traces the development of ideas about the structure of the atom and the arrangement of elements in the modern Periodic Table. Both stories show how scientific theories develop as new evidence is made available that either supports or contradicts current ideas.

Atomic structure is used to help explain the behaviour of the elements. Trends and patterns shown by the physical and chemical properties in groups and in the transition metals are studied.

The first two topics of the chapter give opportunities for learners to develop understanding of ideas about science; how scientific knowledge develops, the relationship between evidence and explanations, and how the scientific community responds to new ideas. The later topics present some of the most important models which underpin an understanding of atoms, chemical behaviour and patterns and how reactions are represented in chemical equations.

Topic C2.1 looks at the development of ideas about the atom and introduces the modern model for atomic structure, including electron arrangements. Topic C2.2 considers the development of the modern Periodic Table and the patterns that exist within it, focusing on Groups 1 and 7, with some reference to Group 0. Topic C2.3 focuses on extending an understanding of atomic structure to explain the ionic bonding between ions in ionic compound. This leads on to Topic C2.4 which studies using equations and symbols to summarise reactions. Finally, in Topic C2.5, separate science only content addresses the unique nature of the transition elements.

Learning about Chemical Patterns before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know the properties of the different states of matter (solid, liquid and gas) in terms of the particle model, including gas pressure
- know changes of state in terms of the particle model
- be aware of a simple (Dalton) atomic model
- know differences between atoms, elements and compounds
- know chemical symbols and formulae for elements and compounds
- know conservation of mass in changes of state and chemical reactions
- understand chemical reactions as the rearrangement of atoms
- be able to represent chemical reactions using formulae and using equations
- know some displacement reactions
- know what catalysts do
- be aware of the principles underpinning the Mendeleev Periodic Table
- know some ideas about the Periodic Table: periods and groups; metals and non-metals
- know how some patterns in reactions can be predicted with reference to the Periodic Table
- know some properties of metals and non-metals.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about chemical patterns at GCSE (9–1)

C2.1 How have our ideas about atoms developed over time?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	
The modern model of the atom developed over time. Stages in the development of the model included ideas by the ancient Greeks (4 element ideas), Dalton (first particle model), Thomson ('plum pudding' model), Rutherford (idea of atomic nucleus) and Bohr (shells of electrons). Models were rejected, modified and extended as new evidence became available. The development of the atomic model involved scientists suggesting explanations, making and checking predictions based on their explanations, and building on each other's work (IaS3). The Periodic Table can be used to find the atomic number and relative atomic mass of an atom of an element, and then work out the numbers of protons, neutrons and electrons. The number of electrons in each shell can be represented by simple conventions such as dots in circles or as a set of numbers (for example, sodium as 2.8.1). Atoms are small – about 10^{-10} m across, and the nucleus is at the centre, about a hundred-thousandth of the diameter of the atom. Molecules are larger, containing from two to hundreds of atoms. Objects that can be seen with the naked eye contain millions of atoms.	1.	describe how and why the atomic model has changed over time to include the main ideas of Dalton, Thomson, Rutherford and Bohr
	2.	describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with most of the mass in the nucleus
	3.	recall relative charges and approximate relative masses of protons, neutrons and electrons
	4.	estimate the size and scale of atoms relative to other particles M1d
	5.	recall the typical size (order of magnitude) of atoms and small molecules
	6.	relate size and scale of atoms to objects in the physical world M1d
	7.	calculate numbers of protons, neutrons and electrons in atoms, given atomic number and mass number of isotopes or by extracting data from the Periodic Table M1a

Linked learning opportunities

Specification links:

- atoms and radiation (P5.1)

Ideas about Science:

- understanding how scientific explanations and models develop in the context of changing ideas about the atomic model (IaS3)